

Seismological Society of America
Technical Sessions
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The following schedule of events and abstracts are valid until 22 March 2022 and subject to change.

The Society is grateful to the SSA 2022 Co-chairs Jackie Caplan-Auerbach, Western Washington University, and David Schmidt, University of Washington.

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Modeling, Collecting and Communicating Post-earthquake Hazard and Impact Information

Oral Session · Wednesday 20 April · 4:30 PM Pacific

Conveners: David J. Wald, U.S. Geological Survey (wald@usgs.gov); Heidi Stenner, Geohazards International (stenner@geohaz.org); Eric Fielding, NASA Jet Propulsion Laboratory (eric.j.fielding@jpl.nasa.gov); Haeyoung Noh, Stanford University (noh@stanford.edu); Susu Xu, SUNY Stonybrook (susu.xu@stonybrook.edu); Kate E. Allstadt, U.S. Geological Survey (kallstadt@usgs.gov)

Enhanced Rapid Earthquake Ground Failure and Impact Estimates With Remotely Sensed and Ground Truth Constraints

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Estimating earthquake impacts using physical or empirical models is challenging because the main components of loss estimation—namely shaking, exposure and vulnerabilities—entail inherent uncertainties. Loss modeling in near-real-time adds additional challenges, yet expectations for actionable information with a reasonable level of confidence in the results are real. Fortunately, advancements in remote sensing, rapid in-situ impact reporting and machine learning—combined with new datasets such as global building footprints—will allow for innovative data-fusion strategies that integrate with existing models and should greatly improve the accuracy and spatial resolution of rapid shaking and loss estimates. We are exploring two approaches.

First, early reports of casualties are used in a Bayesian updating fashion to constrain the possible range of fatalities and lower the prior models' uncertainties (see Engler et al., this meeting). Second, in the form of Damage Proxy Maps, remotely sensed satellite radar data are used in a Bayesian causal graph framework combined with machine learning to optimize the mapping among the physical processes that cause shaking-based building damage, landslides and liquefaction to prior expectation models. The casual graph framework also affords the potential for removing anthropogenic noise contained in the imagery. Our main findings to date are that (1) updating the PAGER fatality model can prevent cases where PAGER losses are initially significantly off (e.g., the incorrect alert level) by quickly allowing updates, and that (2) the imagery—while slower than ground-truth observations—provide more spatially accurate impact assessments, well beyond the capabilities of the generalized loss and ground failure models. Ultimately, our two-fold model updating strategy will accommodate key ground-truth observations such as fatality reports, locations of building damage and ground failure reports to converge on actual losses more rapidly.

